



ATM Software

XFS4IoT SP-Dev Workgroup

1st September 2026

- Our journey over the years
- Guest speakers
- Highlights
- What's next





The workgroup's journey: 2021-2026

- **Kicked off in March 2021**
 - 150+ attendees from 60+ companies
 - Presented how the XFS Committee works
 - Proposed how to work on the SP-Dev framework together
- **Over that year:**
 - Worked on the first XFS4IoT version, 2021-1
 - In July, published competition law compliance guidelines
 - First SP-Dev framework published on GitHub (.NET 5)
 - Supported CardReader, CashDispenser, Common, CashManagement, Crypto, KeyManagement, Keyboard, Pinpad, TextTerminal and Printer frameworks

- XFS4IoT 2021-1 formally published by CEN (now AFNOR XFS)
- SP-Dev Framework evolution
 - Framework updated to .NET 6
 - Added support for BarcodeReader, Biometrics, Auxiliaries, CashAcceptor, CashRecycler, VendorApplication and VendorMode
- Introduced SP templates
 - Used to create new C# projects for SP development

- XFS4IoT 2023-2 submitted to CEN
- Demo
 - KAL application running in the cloud, with a Cash Recycling device



- XFS4IoT 2023-2 formally published by CEN
- SP-Dev framework evolution
 - Framework updated to .NET 8
 - Added support for Camera and CheckScanner
- XFS4IoT 2024-3 submitted to CEN
- Experimented with Android-based devices (POS)

- XFS4IoT 2024-3 formally published by CEN
- SP-Dev framework evolution
 - Added support for German DK and IBNS interface
 - Migration to JSON form and media formats for Printer
- Experimented with
 - Ahead of Time compilation
 - Hybrid XFS3/XFS4IoT systems
 - Running on small footprint boards (Meadow F7)

- Cross-platform support for TLS on Android, Linux and Windows
- Experimented with nano framework on Meadow F7



Guest speakers



- 2021: Red Hat and Serquo
- 2022: Global Payments and Oberthur Cash Protection
- 2023: Sigma and Oberthur Cash Protection
- 2024: First National Bank, Synaforce and SBS
- 2026: Synaforce



Some highlights

- Running a truly cloud-based architecture, with the application running in Azure and the hardware running locally - using the XFS4IoT SP-Dev framework to implement the connection





- Support for true End-to-end security on dispensers, for the first time making it possible to fully control the security of cash, and blocking a host of existing attacks

- 20+ demos
 - How to create SPs
 - How to upgrade projects with newer versions
 - Working SPs with several real devices
 - SPs for Windows, Android, Linux systems
 - Hybrid XFS3/XF4IoT applications
 - and more
- 28 packages / 416,610 downloads

A scenic landscape featuring a two-lane asphalt road with a white dashed center line that curves gently into the distance. The road is flanked by green, grassy hills. In the background, a thick layer of white clouds fills the valley, with the sun setting behind them, creating a vibrant orange and yellow glow across the sky. The overall atmosphere is peaceful and contemplative.

What's next

- Submit XFS4IoT 2026-4 to AFNOR XFS in December
- Framework updates
 - Support for version 2026-4
 - Migration to .NET 11
 - Cash Acceptor End-to-end security
 - And more...

What's next



- We're stopping the SP-Dev workgroup meetings... for now
- All past presentations still available on GitHub
- Demos on KAL's YouTube channel (<https://www.youtube.com/@ATMsoftware>)
- KAL is publishing tutorials
- There will be a newsletter featuring news, how-tos and useful resources

<https://www.kal.com/xfs4iot>





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